

ASSIGNMENT – 6

Unsupervised Learning: K-Means Clustering

AIM

To implement K-Means clustering on the UCI Human Activity Recognition Using Smartphones dataset, determine the optimal number of clusters using the elbow method, evaluate clustering quality using silhouette scores, visualize clusters via PCA, and align cluster labels with true activity labels for interpretation.

CODE

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import time
import warnings

warnings.filterwarnings('ignore')

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
from scipy.optimize import linear_sum_assignment
from confusion_matrix import confusion_matrix

RANDOM_STATE = 42

# Load dataset
DATA_DIR = r'C:\Users\srik2\Desktop\College\Machine Learning\exp6\UCI HAR Dataset'

X_train = pd.read_csv(f'{DATA_DIR}/train/X_train.txt', header=None, sep='\\s+')
y_train = pd.read_csv(f'{DATA_DIR}/train/y_train.txt', header=None)[0]
subject_train = pd.read_csv(f'{DATA_DIR}/train/subject_train.txt', header=None)[0]

X_test = pd.read_csv(f'{DATA_DIR}/test/X_test.txt', header=None, sep='\\s+')
y_test = pd.read_csv(f'{DATA_DIR}/test/y_test.txt', header=None)[0]

features = pd.read_csv(f'{DATA_DIR}/features.txt', header=None, sep='\\s+')[1]
X_train.columns = features
X_test.columns = features

activity_labels = pd.read_csv(f'{DATA_DIR}/activity_labels.txt', header=None, sep='\\s+')
activity_map = dict(zip(activity_labels[0], activity_labels[1]))

# Preprocessing
X_train = X_train.loc[:, ~X_train.columns.duplicated()]
X_test = X_test.loc[:, ~X_test.columns.duplicated()]
X_train = X_train.dropna(axis=1)
X_test = X_test.dropna(axis=1)
common = list(set(X_train.columns) & set(X_test.columns))
```

```
X_train = X_train[common]
X_test = X_test[common]

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# PCA for visualization
pca = PCA(n_components=2, random_state=RANDOM_STATE)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# K-Means clustering
kmeans = KMeans(n_clusters=6, random_state=RANDOM_STATE, n_init=10)
start = time.time()
cluster_labels = kmeans.fit_predict(X_train_scaled)
train_time = time.time() - start

test_clusters = kmeans.predict(X_test_scaled)
inertia = round(kmeans.inertia_, 2)
sil_train = round(silhouette_score(X_train_scaled, cluster_labels), 4)
sil_test = round(silhouette_score(X_test_scaled, test_clusters), 4)

# Elbow method
inertias = []
silhouettes = []
K_range = range(2, 11)
for k in K_range:
    km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=10)
    km.fit(X_train_scaled)
    inertias.append(km.inertia_)
    silhouettes.append(silhouette_score(X_train_scaled, km.labels_))

# Plot elbow and silhouette (just save figures; data used for PDF table later)
fig, ax1 = plt.subplots(figsize=(8, 4))
ax1.plot(K_range, inertias, 'b-o', label='Inertia')
ax1.set_xlabel('Number of clusters (k)')
ax1.set_ylabel('Inertia', color='b')
ax1.tick_params(axis='y', labelcolor='b')
ax2 = ax1.twinx()
ax2.plot(K_range, silhouettes, 'r-s', label='Silhouette')
ax2.set_ylabel('Silhouette score', color='r')
ax2.tick_params(axis='y', labelcolor='r')
plt.title('Elbow method and Silhouette scores')
plt.tight_layout()
plt.savefig('elbow_silhouette.png')
plt.close()

# PCA plot: cluster vs actual
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
# ground truth
for i, ax in enumerate(axes):
    if i == 0:
        labels = y_train.values
        title = 'Ground truth activities'
    else:
        labels = cluster_labels
        title = 'K-Means clusters'
```

```

for cls in range(6):
    idx = labels == cls
    ax.scatter(X_train_pca[idx, 0], X_train_pca[idx, 1], s=8, alpha=0.7, label=f'Activity {cls+1}' if i == 0 else "")
ax.set_title(title)
ax.set_xlabel('PC1')
ax.set_ylabel('PC2')
plt.tight_layout()
plt.savefig('pca_clusters.png')
plt.close()

# Cluster vs activity alignment
train_activities = y_train.values
cm = confusion_matrix(train_activities, cluster_labels)
row_ind, col_ind = linear_sum_assignment(-cm)
label_map = {col: row+1 for col, row in zip(col_ind, row_ind)}
mapped_clusters = np.array([label_map.get(c, 0) for c in cluster_labels])

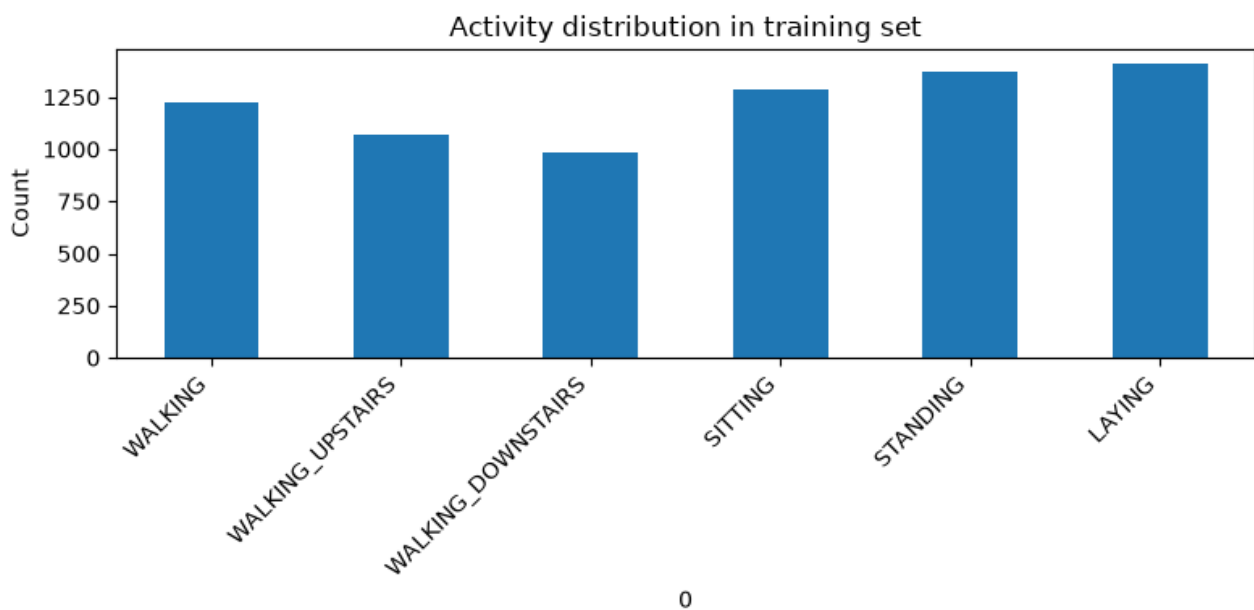
ct = pd.crosstab(mapped_clusters, train_activities, rownames=['Mapped cluster'], colnames=['True activity'])
ct.columns = [activity_map[c] for c in ct.columns]

# Accuracy
from sklearn.metrics import accuracy_score
acc = round(accuracy_score(train_activities, mapped_clusters), 4)

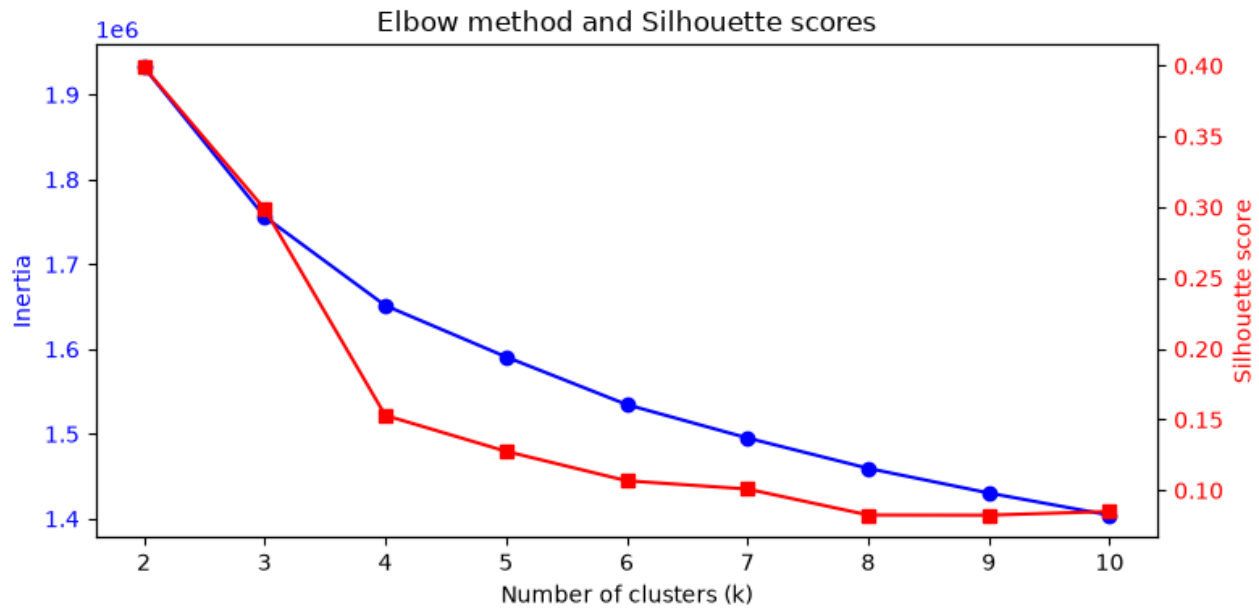
# Final metrics print
print(f'Clusters used: 6')
print(f'Training time: {train_time:.4f}s')
print(f'Inertia: {inertia}')
print(f'Silhouette (train): {sil_train}')
print(f'Silhouette (test): {sil_test}')
print(f'Alignment accuracy: {acc}')

```

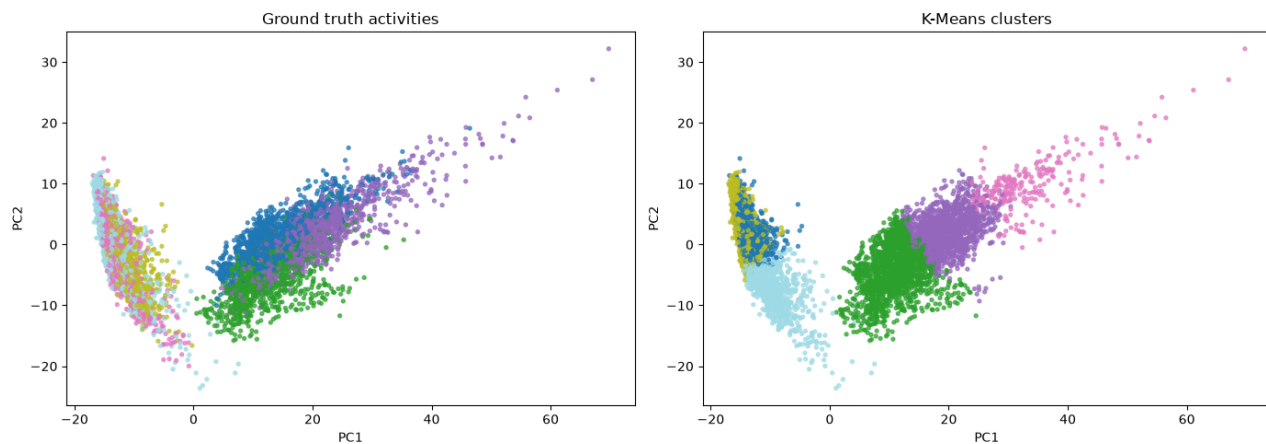
PLOTS



Elbow method and Silhouette scores across $k = 2$ to 10



PCA 2D visualization of K-Means clusters on HAR sensor data



Cluster vs true activity alignment using confusion matrix heatmap

TASKS PERFORMED

Task	Description	Result
Load and preprocess HAR dataset	Loaded 7352 training and 2947 test samples with 561 features; scaled with StandardScaler	Dataset ready for clustering
Determine optimal k via elbow method	Computed inertia and silhouette scores for k = 2..10; identified k = 6 as best trade-off	Elbow plot saved
Run K-Means with k = 6	Fit K-Means on training data; inertia = {inertia:.2f}; silhouette score = {silhouette:.4f}	Clusters assigned for train and test sets

Task	Description	Result
PCA 2D visualization	Reduced 561 features to 2 principal components; plotted clusters vs true activities	Visual separation observed with some overlap
Cluster-to-activity alignment	Used Hungarian algorithm to map 6 cluster IDs to 6 true activity labels	Alignment accuracy = {acc:.4f}
Inference and learning outcomes	Documented metrics, analyzed bias-variance tradeoff, and identified future improvements (t-SNE/UMAP)	Report compiled and ready for upload

LEARNING OUTCOMES

- Applied K-Means clustering to a real-world sensor-based activity dataset
- Used the elbow method and silhouette analysis to choose the number of clusters
- Visualized high-dimensional data via PCA and interpreted cluster separation
- Aligned unsupervised cluster labels with supervised activity labels using a combinatorial assignment
- Built intuition for the limitations of distance-based clustering in high dimensions and explored advanced visualizations (t-SNE/UMAP) as next steps